TECA - Toolkit for Extreme Climate Analysis

A parallel toolkit for detecting extreme events in large climate datasets

Tempest TC Detect

Tropical cyclone detection algorithm based on *TempestExtremes*

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Overview

- Porting the ***DetectNodes*** and ***StitchNodes*** functions from ***TempestExtremes*** to **TECA**: CPU version
- The GPU version of ***TECA Tempest TC Detect*** was implemented
- How it works and is parallelized
- Both CPU and GPU versions were validated
- How to use ***TECA Tempest TC Detect*** vs ***TempestExtremes***

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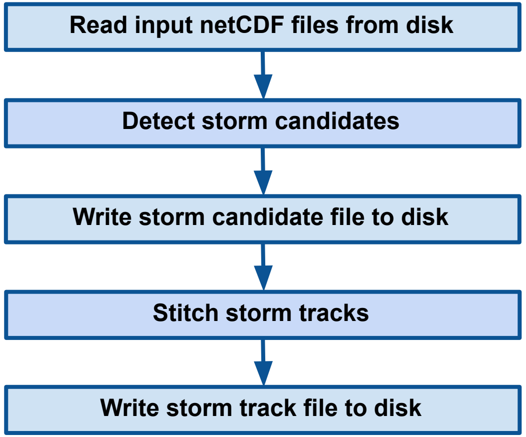
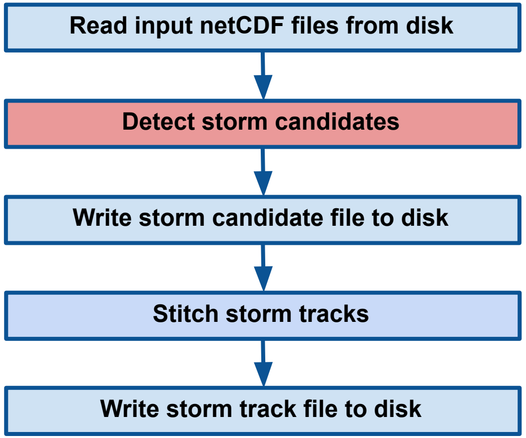
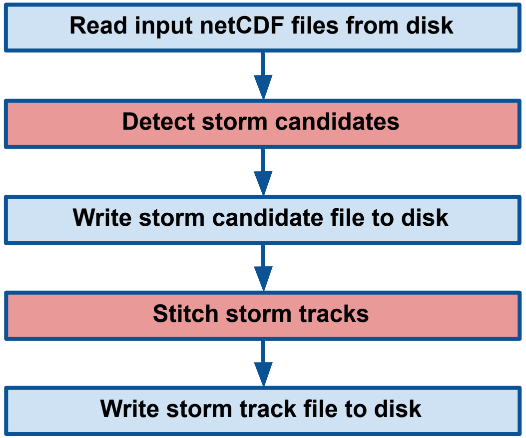
TECA Tempest TC Detect



TECA Tempest TC Detect

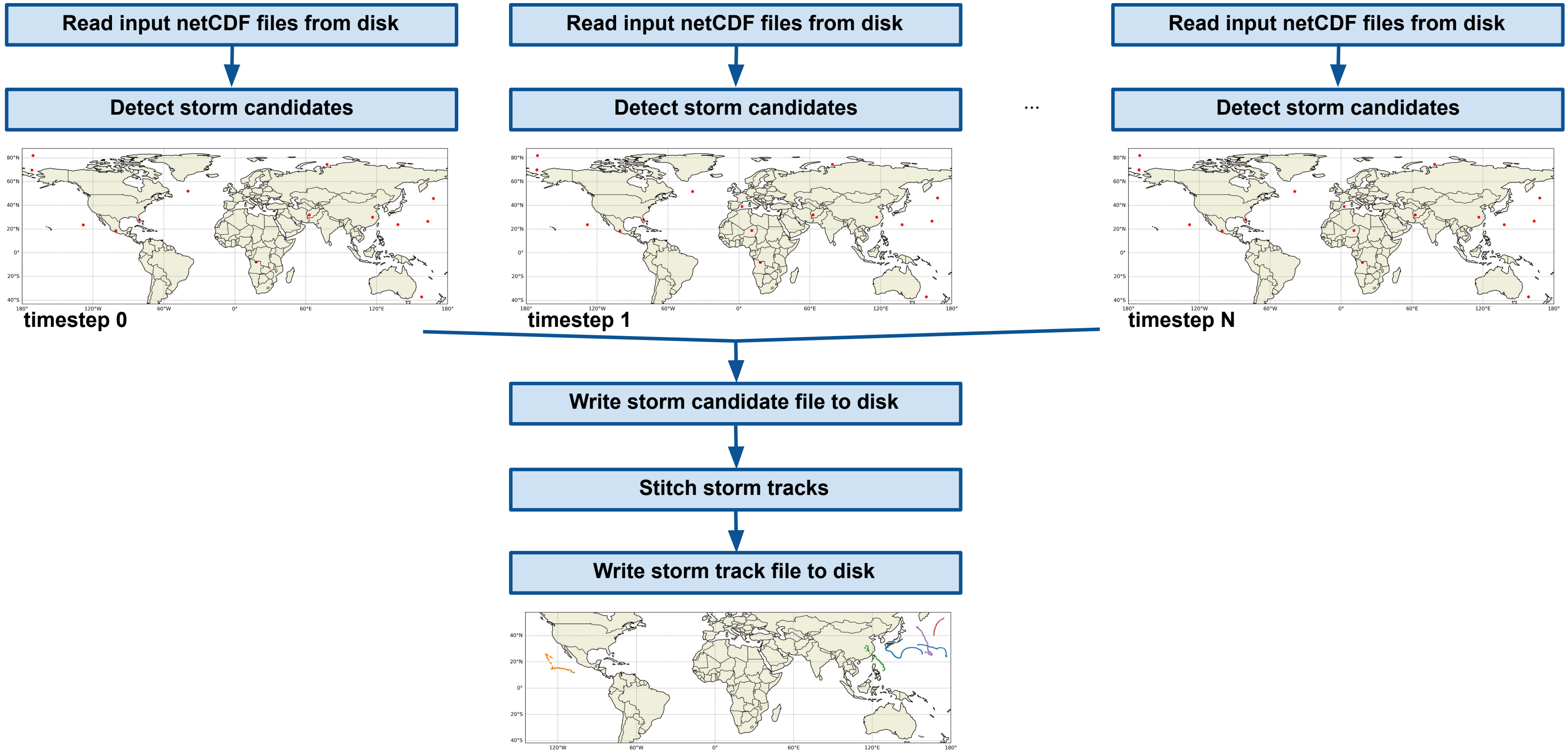


TECA Tempest TC Detect

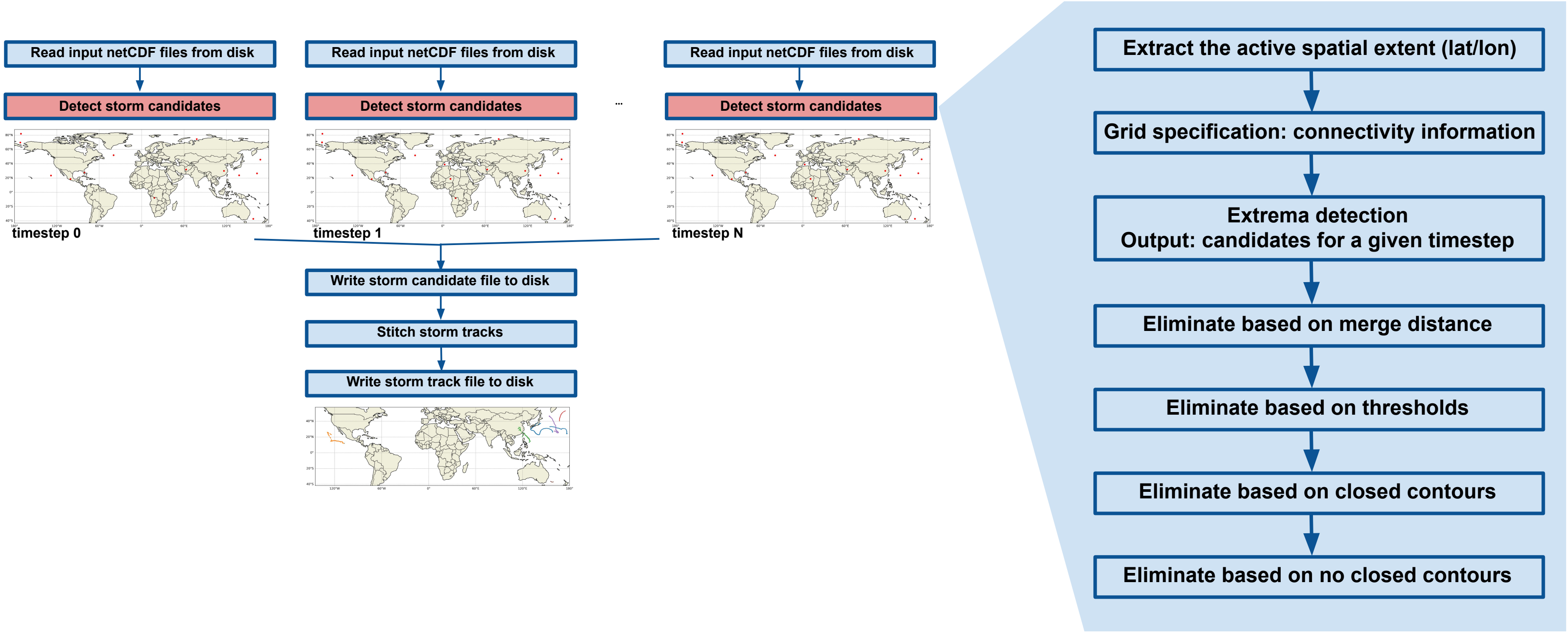


TECA Tempest TC Detect

Parallelization: distribute timesteps (frames) among MPI ranks (CPU cores and/or GPUs)



TECA Tempest TC Detect

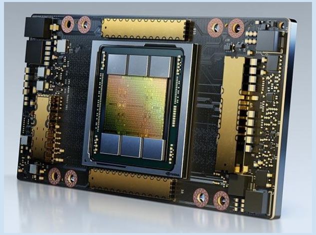


TECA Tempest TC Detect: GPU version

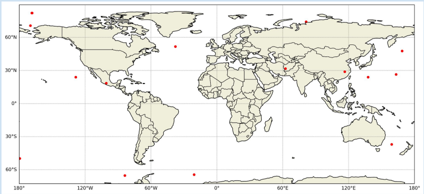
- Read input netCDF files from disk
- Detect storm candidates
- Write storm candidate file to disk
- Stitch storm tracks
- Write storm track file to disk

Detect storm candidates:

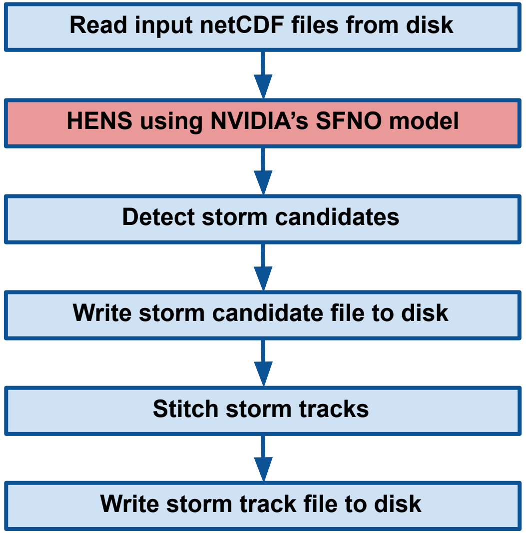
- Extract the active spatial extent (lat/lon)
- Grid specification: connectivity information
- Extrema detection
  Output: candidates for a given timestep
- Eliminate based on merge distance
- Eliminate based on thresholds
- Eliminate based on closed contours
- Eliminate based on no closed contours



Identify local extrema by comparing against all neighboring grid points for a given time step (e.g. 00h on Aug 1, 2017)



TECA Tempest TC Detect + TECA-HENS



An example workflow of TECA driving NVIDIA’s SFNO simulation and performing inline TC detection on HENS output data available for use on the GPU by SFNO simulation.

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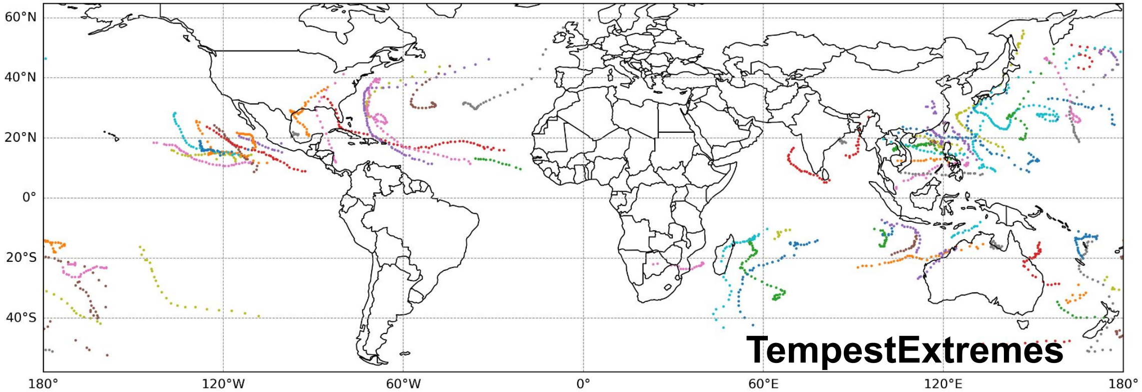
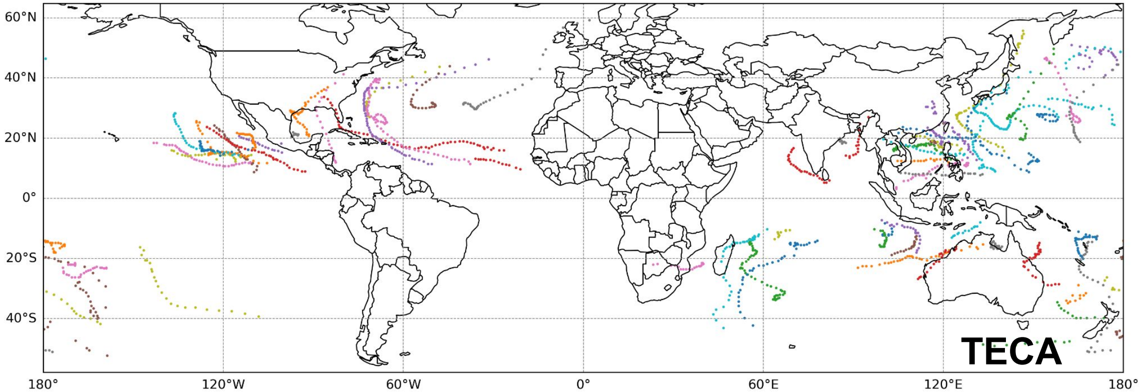
Tracks 2017





Problem: tropical cyclone tracking

Computational Resources: 1 PM-CPU node
64 MPI ranks

Data: ERA5 reanalysis data for the year 2017
721 latitude x 1440 longitude grid points
6 hourly time resolution (1,460 timesteps)

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TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
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  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
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  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes: files_IN.txt

```
e5.oper.an.pl.128_129_z.ll025sc.2017010100_2017010123.nc;e5.oper.invariant.128_129_zs.ll025sc.2017010100_2017013123_c.nc;
e5.oper.an.sfc.128_151_msl.ll025sc.2017010100_2017013123.nc;e5.oper.an.sfc.128_166_10v.ll025sc.2017010100_2017013123.nc;e
5.oper.an.sfc.128_165_10u.ll025sc.2017010100_2017013123.nc
e5.oper.an.pl.128_129_z.ll025sc.2017010200_2017010223.nc;e5.oper.invariant.128_129_zs.ll025sc.2017010100_2017013123_c.nc;
e5.oper.an.sfc.128_151_msl.ll025sc.2017010100_2017013123.nc;e5.oper.an.sfc.128_166_10v.ll025sc.2017010100_2017013123.nc;e
5.oper.an.sfc.128_165_10u.ll025sc.2017010100_2017013123.nc
e5.oper.an.pl.128_129_z.ll025sc.2017010300_2017010323.nc;e5.oper.invariant.128_129_zs.ll025sc.2017010100_2017013123_c.nc;
e5.oper.an.sfc.128_151_msl.ll025sc.2017010100_2017013123.nc;e5.oper.an.sfc.128_166_10v.ll025sc.2017010100_2017013123.nc;e
5.oper.an.sfc/201701/e5.oper.an.sfc.128_165_10u.ll025sc.2017010100_2017013123.nc
…
```

TECA: files_IN.mcf

```
data_root = /pscratch/sd/a/asdufek/projects/tc_detector_experiments/links_2017

z_axis_variable = level
clamp_dimensions_of_one = 1

[cf_reader]
variables = Z
regex = %data_root%/e5.oper.an.pl.128_129_z.ll025sc.*\.nc
provides_time
provides_geometry

[cf_reader]
variables = MSL
regex = %data_root%/e5.oper.an.sfc.128_151_msl.ll025sc.*\.nc
z_axis_variable = ""

[cf_reader]
variables = VAR_10U
regex = %data_root%/e5.oper.an.sfc.128_165_10u.ll025sc.*\.nc
z_axis_variable = ""

[cf_reader]
variables = VAR_10V
regex = %data_root%/e5.oper.an.sfc.128_166_10v.ll025sc.*\.nc
z_axis_variable = ""

[cf_reader]
variables = ZS
regex = %data_root%/../e5.oper.invariant.128_129_zs.ll025sc.1979010100_1979010100.nc
z_axis_variable = ""
```

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  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
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  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
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  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes: TC_tracks.txt

```
start 13 2017  1  1  0
   733   472   183.250000  -28.000000  1.002096e+05   1.297175e+01   -1.353027e+00  2017  1  1  0
   733   472   183.250000  -28.000000  1.002019e+05   1.318713e+01   -1.353027e+00  2017  1  1  1
   734   472   183.500000  -28.000000  1.001956e+05   1.267869e+01   2.485352e-01   2017  1  1  2
   734   473   183.500000  -28.250000  1.001489e+05   1.260013e+01   -6.850586e-01  2017  1  1  3
   734   473   183.500000  -28.250000  1.001776e+05   1.276967e+01   -6.850586e-01  2017  1  1  4
   735   473   183.750000  -28.250000  1.002030e+05   1.234732e+01   -6.108398e-01  2017  1  1  5
   735   473   183.750000  -28.250000  1.002892e+05   1.194708e+01   -6.108398e-01  2017  1  1  6
   735   473   183.750000  -28.250000  1.003377e+05   1.131666e+01   -6.108398e-01  2017  1  1  7
   736   473   184.000000  -28.250000  1.004116e+05   1.014044e+01   1.041504e+00   2017  1  1  8
   737   474   184.250000  -28.500000  1.004900e+05   1.002437e+01   -1.556152e+00  2017  1  1  9
   737   474   184.250000  -28.500000  1.006181e+05   9.321583e+00   -1.556152e+00  2017  1  1  10
   738   475   184.500000  -28.750000  1.006052e+05   9.689748e+00   -3.256836e-01  2017  1  1  11
   738   475   184.500000  -28.750000  1.005904e+05   1.041747e+01   -3.256836e-01  2017  1  1  12
```

TECA: TC_tracks.csv

```
# teca_table v1
# 5.0.0-335-gdc45879
# calendar = "gregorian"
# time_units = "hours since 1900-01-01 00:00:00"
"storm_id(7)", "path_length(7)", "year(3)", "month(3)", "day(3)", "hour(3)", "i(3)", "j(3)", "lat(12)", "lon(12)", "MSL(12)",
"surface_wind(12)", "ZS(12)"
0, 13, 2017, 1, 1, 0, 733, 248, -2.800000000000000000e+01, 1.832500000000000000e+02, 1.002096000000000582e+05,
1.297175000000000114e+01, -1.353026999999999797e+00
0, 13, 2017, 1, 1, 1, 733, 248, -2.800000000000000000e+01, 1.832500000000000000e+02, 1.002018999999999418e+05,
1.318712999999999797e+01, -1.353026999999999797e+00
0, 13, 2017, 1, 1, 2, 734, 248, -2.800000000000000000e+01, 1.835000000000000000e+02, 1.001956000000000582e+05,
1.267868999999999571e+01, 2.485350000000000592e-01
0, 13, 2017, 1, 1, 3, 734, 247, -2.825000000000000000e+01, 1.835000000000000000e+02, 1.001488999999999418e+05,
1.260013000000000052e+01, -6.850589999999997352e-01
0, 13, 2017, 1, 1, 4, 734, 247, -2.825000000000000000e+01, 1.835000000000000000e+02, 1.001776000000000582e+05,
1.276966999999999632e+01, -6.850589999999997352e-01
0, 13, 2017, 1, 1, 5, 735, 247, -2.825000000000000000e+01, 1.837500000000000000e+02, 1.002030000000000000e+05,
1.234731999999999851e+01, -6.108400000000004960e-01
0, 13, 2017, 1, 1, 6, 735, 247, -2.825000000000000000e+01, 1.837500000000000000e+02, 1.002891999999999709e+05,
1.194707999999999700e+01, -6.108400000000004960e-01
0, 13, 2017, 1, 1, 7, 735, 247, -2.825000000000000000e+01, 1.837500000000000000e+02, 1.003376999999999709e+05,
1.131666000000000608e+01, -6.108400000000004960e-01
0, 13, 2017, 1, 1, 8, 736, 247, -2.825000000000000000e+01, 1.840000000000000000e+02, 1.004116000000000582e+05,
1.014043999999999898e+01, 1.041503999999999853e+00
0, 13, 2017, 1, 1, 9, 737, 246, -2.850000000000000000e+01, 1.842500000000000000e+02, 1.004900000000000000e+05,
1.002436999999999337e+01, -1.556151999999999797e+00
0, 13, 2017, 1, 1, 10, 737, 246, -2.850000000000000000e+01, 1.842500000000000000e+02, 1.006181000000000582e+05,
9.321583000000003963e+00, -1.556151999999999797e+00
0, 13, 2017, 1, 1, 11, 738, 245, -2.875000000000000000e+01, 1.845000000000000000e+02, 1.006051999999999709e+05,
9.689747999999998057e+00, -3.256839999999997352e-01
0, 13, 2017, 1, 1, 12, 738, 245, -2.875000000000000000e+01, 1.845000000000000000e+02, 1.005903999999999418e+05,
1.041746999999999786e+01, -3.256839999999997352e-01
```

TempestExtremes:

```
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  --timefilter "6hr" \
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  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
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srun -n 1 StitchNodes \
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  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

TECA:

```
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  --candidate_file files_DN.csv \
  --time_filter 6 \
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  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
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  --surface_wind_u VAR_10U \
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  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
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  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

_ABS()
_SIGN()
_SUM()
_AVG()
_PROD()
_DIV()
_SQRT()
_MEAN()
…

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --300mb_height Z300 --500mb_height Z500 \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0; VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

Name of x and y coordinate variables

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

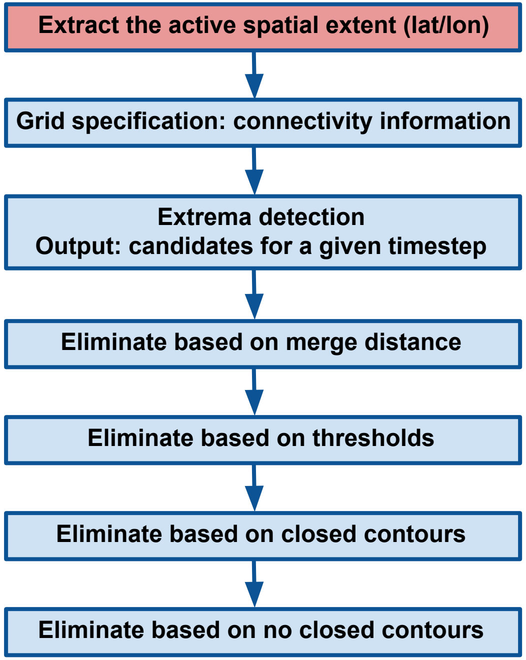
srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

The maximum/minimum latitude/longitude for candidate points.

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x axis variable longitude --y axis variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

Detect storm candidates



TempestExtremes:

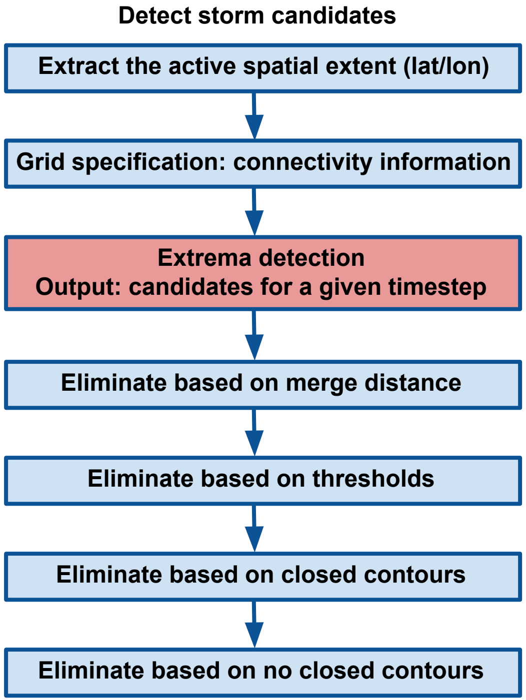
```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

The input variable to use for initially selecting local minima.

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```



TempestExtremes:

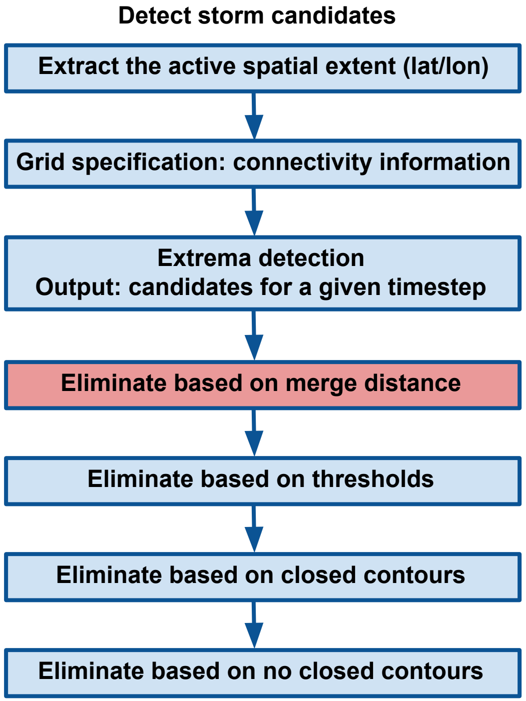
```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

Candidate points with a distance (in degrees) shorter than the specified value are merged.

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```



TempestExtremes:

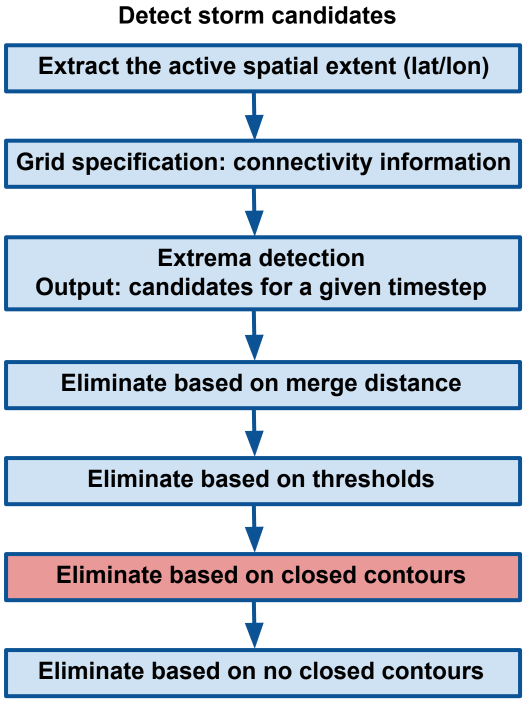
```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

if any paths exist from the candidate point that reach the specified distance before achieving the specified delta then we say no closed contour is present.

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```



TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0; DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

A comma-separated list of names of the auxiliary columns within the candidates output file

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

The maximum distance between candidates along a path (in great-circle degrees).

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

The minimum length of a path

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

The number of allowed missing points between spatially proximal candidate points while still considering them part of the same path.

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TempestExtremes:

```
srun -n 64 DetectNodes \
  --in_data_list files_IN.txt \
  --out_file_list files_DN.txt \
  --timefilter "6hr" \
  --closedcontourcmd "MSL,200.0,5.5,0;_DIFF(Z(300hPa),Z(500hPa)),-58.8,6.5,1.0" \
  --outputcmd "MSL,min,0;_VECMAG(VAR_10U,VAR_10V),max,2;ZS,min,0" \
  --latname "latitude" --lonname "longitude" \
  --maxlat 90 --minlat -90 --maxlon 359.75 --minlon 0 \
  --searchbymin MSL --mergedist 6.0

srun -n 1 StitchNodes \
  --in_list files_DN.txt \
  --out TC_tracks.txt \
  --in_fmt "lon,lat,MSL,wind,ZS" \
  --threshold "wind,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --mintime "10" --maxgap "3"
```

Filter paths based on the number of times where a particular threshold is satisfied.

TECA:

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TECA: long command line

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TECA: short command line using default configuration

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --time_filter 6 \
  --x_axis_variable longitude --y_axis_variable latitude \
  --sea_level_pressure MSL \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --geopotential_at_surface ZS
```

TECA: long command line

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --candidate_file files_DN.csv \
  --time_filter 6 \
  --closed_contour_cmd "MSL,200.0,5.5,0;thickness,-58.8,6.5,1.0" \
  --output_cmd "MSL,min,0;surface_wind_speed,max,2;ZS,min,0" \
  --x_axis_variable longitude --y_axis_variable latitude \
  --max_lat 90 --min_lat -90 --max_lon 359.75 --min_lon 0 \
  --search_by_min MSL --merge_dist 6.0 \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --track_file TC_tracks.csv \
  --in_fmt "i,j,lat,lon,MSL,surface_wind_speed,ZS" \
  --track_threshold_cmd "surface_wind_speed,>=,10.0,10;lat,<=,50.0,10;lat,>=,-50.0,10;ZS,<=,15.0,10" \
  --range 8.0 --min_time "10" --max_gap "3"
```

TECA: short command line using default configuration

```
srun -n 64 teca_tempest_tc_detect \
  --input_file files_IN.mcf \
  --time_filter 6 \
  --x_axis_variable longitude --y_axis_variable latitude \
  --sea_level_pressure MSL \
  --surface_wind_u VAR_10U \
  --surface_wind_v VAR_10V \
  --geopotential Z \
  --geopotential_at_surface ZS
```

Next steps

- *TECA Tempest TC Detect* will be made available for use on Perlmutter soon.

- We would appreciate any feedback for improvements, specially regarding the parameter usage and default values.

- We are continually working to improve the performance of *TECA Tempest TC Detect*.